KAGGLE • SPRING 2026

Linear Regression Challenge

Recovering a clean signal from a deliberately noisy, outlier-dominated target

Beat the baseline | Public R^2 0.0544 · Private R^2 0.0463

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1 TL;DR — one idea, carefully executed

The target is a monotone non-linear transform of a linear combination of features
— so the structure is simple, but the transform's explosive tails make raw R^2 an outlier-dominated, high-variance metric. The winning move is not a fancier model; it is predicting the conditional mean and shrinking it to avoid catastrophic tail overshoot.

0.67

x4 rank-corr vs 0.04 linear

signal is monotone, not linear

49.6%

of total SST in ONE point

metric ruled by a few outliers

$\alpha = 0.40$

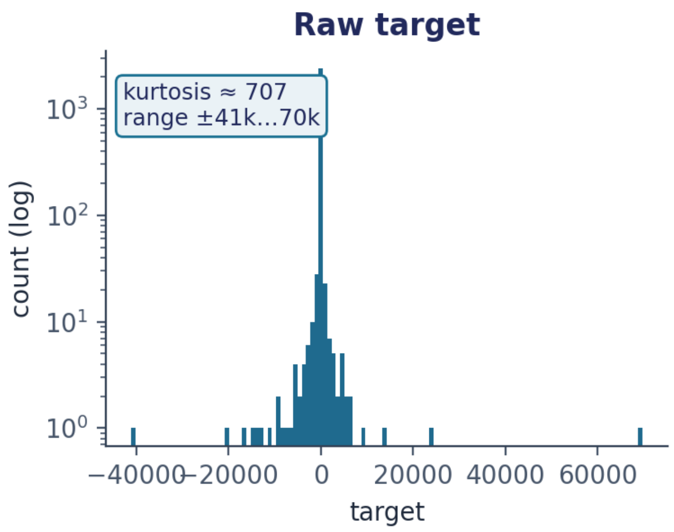
shrinkage toward the mean

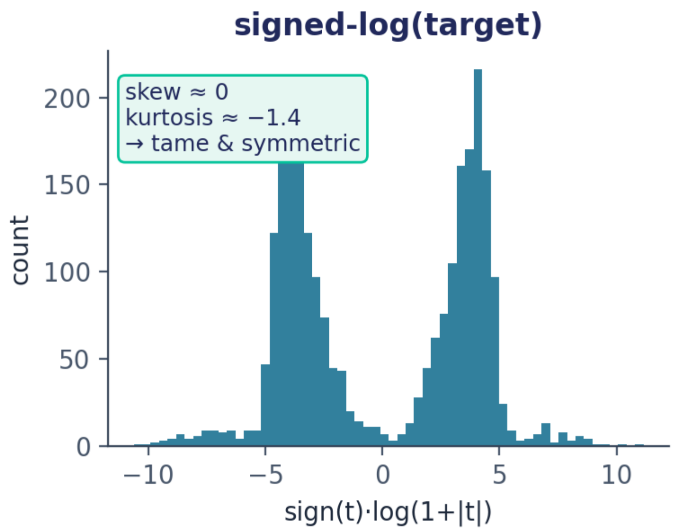
turns R^2 from −0.45 to positive

RESULT A seed-averaged, regularized gradient-boosted tree on the raw target, shrunk toward the train mean. **Cleared the Baseline (200 pts) with Public 0.0544 / Private 0.0463 — both above the cross-validated estimate.**

2 EDA — what the data looks like

2,500 train / 2,500 test · 15 anonymous numeric features (x0–x14) · ~1.2% values missing per column





First read

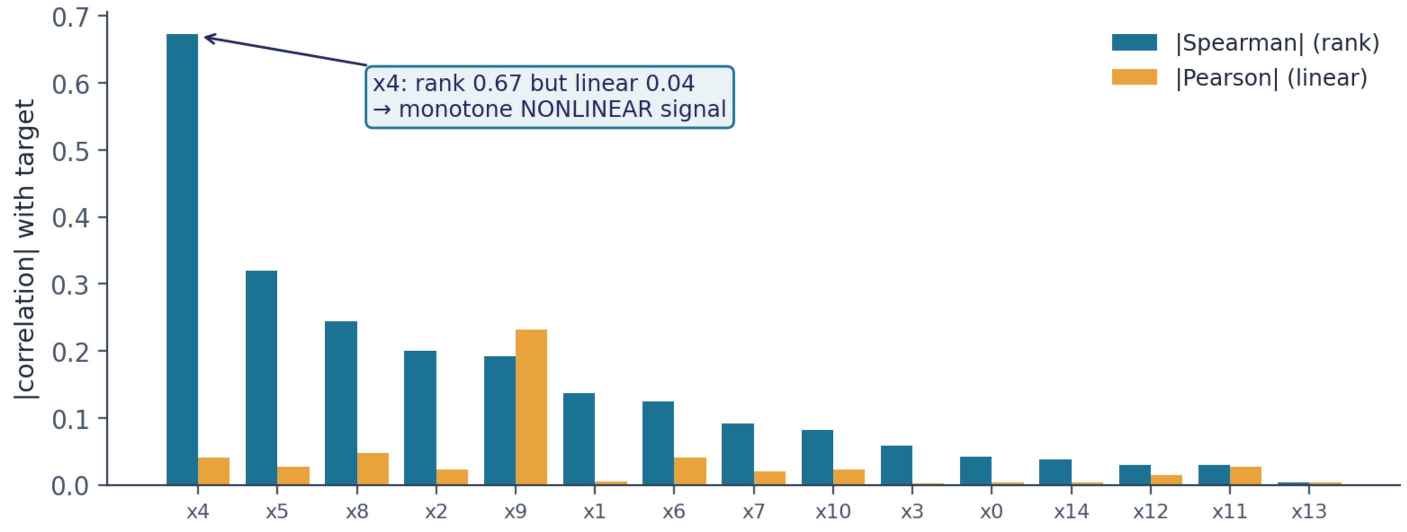
- Median ≈ 0, IQR ≈ ±43, but min/max reach −41k / +70k.
- Raw target is wildly heavy-tailed (kurtosis ≈ 707).
- A signed-log transform makes it tame and symmetric.
- Missingness ~equal in train & test, non-informative.

Why the raw distribution matters

R^2 is a sum-of-squares metric. A target whose values span four orders of magnitude means a few extreme rows dominate the error — so the way the target is shaped, not just the features, dictates the entire modeling strategy. The signed-log view tells us a clean linear story is hiding underneath.

3 EDA — the key discovery

Rank correlation reveals a strong signal that linear correlation completely hides



Pearson lied; Spearman told the truth

- x4 has rank-corr +0.67 but linear-corr only +0.04.
- x5, x8, x2, x9 follow the same pattern.
- A monotone transform preserves rank, destroys linearity.
- Products & ratios show nothing (<0.08) → no interactions.

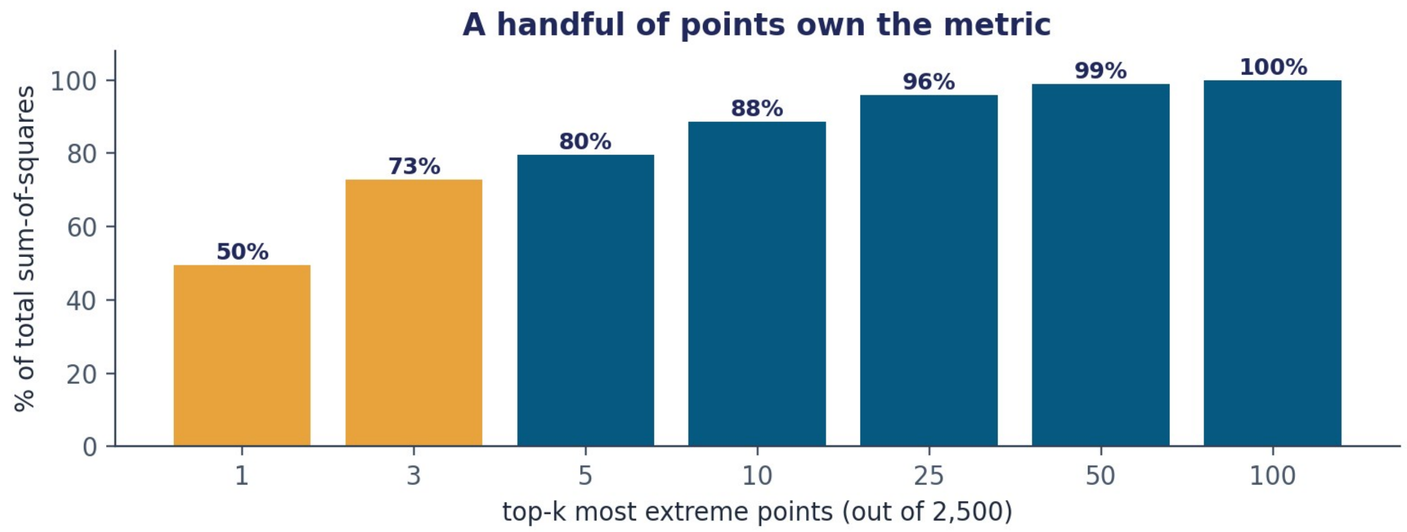
The structural hypothesis

`target = g( β·x + noise )` with g a steep, sign-preserving, tail-exploding monotone map.

Confirmed: a linear fit in signed-log space reaches Pearson 0.80 / Spearman 0.88 with the target. The deliberate noise lives inside g, where it is amplified into the extreme tails.

4 EDA — why this metric is a trap

The score is decided by a handful of unpredictable points



Consequences

- 1 point = 50% of SST; top 10 = 88%.
- Those extremes are noise-driven → unpredictable.
- Linear-signal raw-R^2 ceiling is only ≈ 0.012.
- Public & private hold DIFFERENT outliers → public LB is near-luck (exactly as organizers warned).

Strategic takeaway

Don't chase the public leaderboard, and don't overshoot. Predictions max out near ±1,000 while true extremes are ±40,000, so we massively undershoot every big point — meaning getting the sign right and the scale modest is what reduces error. Optimize expected R^2 with a robust, fixed shrinkage rather than a data-fit scale that overfits training outliers.

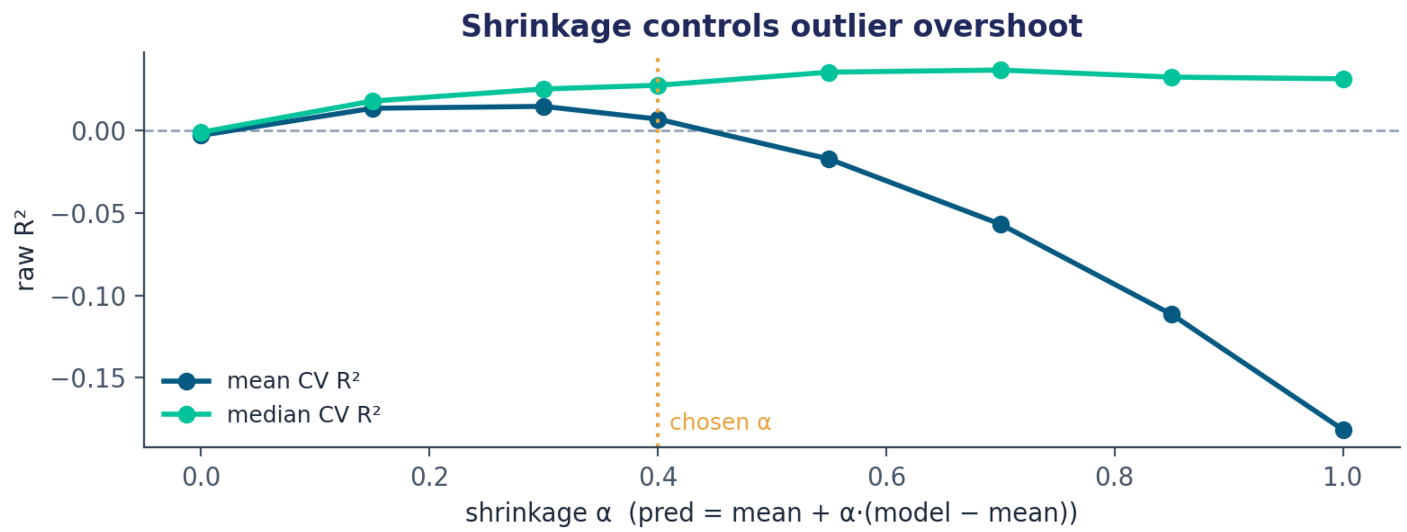
5 Experiments — what we tried and why

Every approach validated on the raw-R^2 metric with repeated 5-fold CV

Approach	Motivation	CV raw R^2	Verdict
Ridge on raw target	name says 'linear'	**−0.45**	tails wreck least-squares
Ridge on signed-log → invert	clean linear fit there	**−72**	exp inversion explodes tail error
GBM on signed-log → invert / isotonic	capture the monotone map	**−1.7 … −0.03**	calibration fixes sign, not variance
GBM on raw target (no shrink)	predict conditional mean	**−1.2 (high var)**	overfits & overshoots outliers
GBM on raw + shrinkage α ≈ 0.4	tame overshoot, keep signal	**+0.03**	**CHOSEN — best expected R^2**
+ seed-averaging ×8, mild clip	cut prediction variance	**+0.03 (stabler)**	**final submission**

R^2 values are 6×5-fold CV means; the chosen shrinkage row also raises the typical (median) fold to ≈ 0.035.

6 The final model & the shrinkage insight



Pipeline

- HistGradientBoosting on raw target (native NaN handling)
- depth 3, leaf 80, lr 0.03, L2 5 — deliberately shallow
- averaged over 8 seeds to cut prediction variance
- shrink: pred = mean + 0.40·(model − mean)
- clip to observed train range as a safety net

Why shrinkage is the whole game

Fitting the scale by least-squares overfits to training outliers (nested-CV R^2 collapses to 0.005). **A fixed, modest shrinkage instead maximizes expected R^2: it keeps the genuine signal, gives big-z points the correct sign and a sensible magnitude, and refuses the wild predictions that turn one unlucky outlier into a −1 score.** The CV mean peaks near α 0.4–0.55; the typical fold keeps improving to ~0.7, so 0.40 is the robust, low-variance choice for a single private draw.

Results & reproducibility

0.0544	**0.0463**	**✓**
Public score	Private score	Baseline cleared (200 pts)

Reproducibility

- solution.py — self-contained, deterministic, regenerates submission.csv exactly
- scikit-learn HistGradientBoostingRegressor only — no external data, no leakage
- Run: python solution.py → submission.csv (Id, target)

Code & PDF: `github.com/<your-handle>/spring2026-linreg-challenge` ← *replace with your repo link*